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```
clear;
clc;
close all;
```

Load Talley's test files

```
sparkData = load('SPARK_ONLY_001.mat');
noiseData = load('WN_ONLY_001.mat');
```

```
sparkIQ = sparkData.eventIQ;
noiseIQ = noiseData.eventIQ;
```

```
fs = sparkData.fs;
fc = sparkData.fc;
```

Basic info

```
fprintf('Sample rate: %.0f Hz\n', fs);
fprintf('Center frequency: %.1f MHz\n\n', fc/1e6);
```

```
Sample rate: 2400000 Hz
Center frequency: 145.9 MHz
```

Magnitude

```
sparkMag = abs(sparkIQ);
noiseMag = abs(noiseIQ);
```

Smooth both signals

```
windowLength = 200;
```

```
sparkSmooth = movmean(sparkMag, windowLength);
noiseSmooth = movmean(noiseMag, windowLength);
```

Measurements - Spark

```
sparkMean = mean(sparkMag);
sparkRMS = rms(sparkMag);
sparkPeak = max(sparkMag);
sparkPeakRMS = sparkPeak / sparkRMS;
```

Measurements - Noise

```
noiseMean = mean(noiseMag);
noiseRMS = rms(noiseMag);
noisePeak = max(noiseMag);
noisePeakRMS = noisePeak / noiseRMS;
```

Print results

```
fprintf('===== SPARK ONLY =====\n');
fprintf('Mean magnitude: %.6f\n', sparkMean);
fprintf('RMS magnitude: %.6f\n', sparkRMS);
fprintf('Peak magnitude: %.6f\n', sparkPeak);
fprintf('Peak / RMS ratio: %.2f\n\n', sparkPeakRMS);

fprintf('===== WHITE NOISE ONLY =====\n');
fprintf('Mean magnitude: %.6f\n', noiseMean);
fprintf('RMS magnitude: %.6f\n', noiseRMS);
fprintf('Peak magnitude: %.6f\n', noisePeak);
fprintf('Peak / RMS ratio: %.2f\n\n', noisePeakRMS);
```

```
===== SPARK ONLY =====
Mean magnitude: 0.000693
RMS magnitude: 0.008422
Peak magnitude: 0.251189
Peak / RMS ratio: 29.83
```

```
===== WHITE NOISE ONLY =====
Mean magnitude: 0.015762
RMS magnitude: 0.017786
Peak magnitude: 0.071870
Peak / RMS ratio: 4.04
```

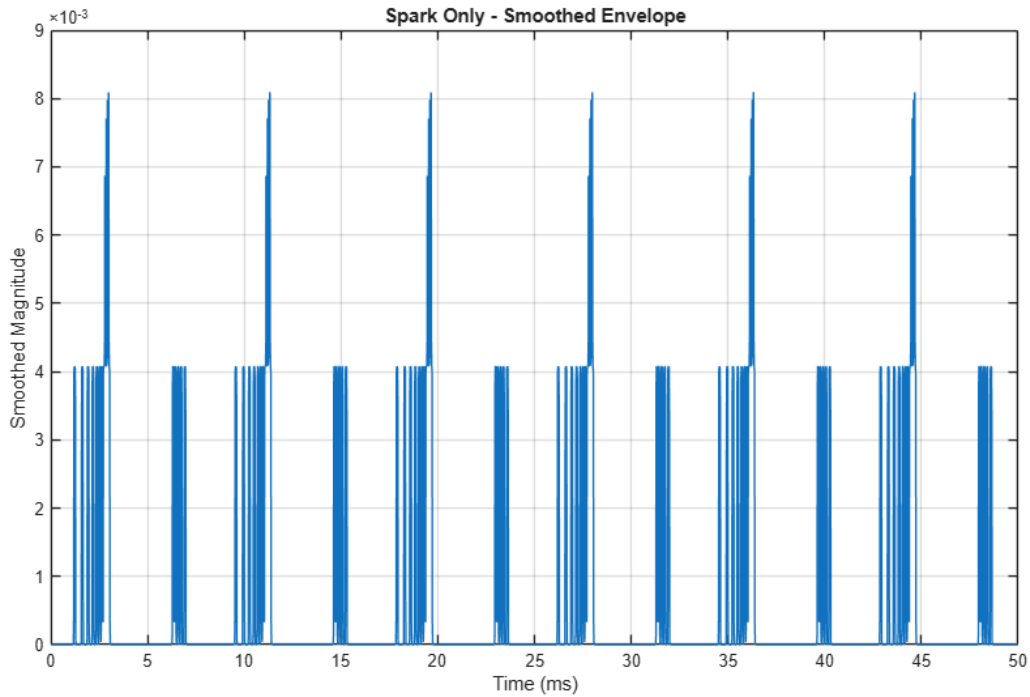
Show first 50 ms

```
showTime = 0.05;
Nshow = round(showTime * fs);

t = (0:length(sparkIQ)-1) / fs;
```

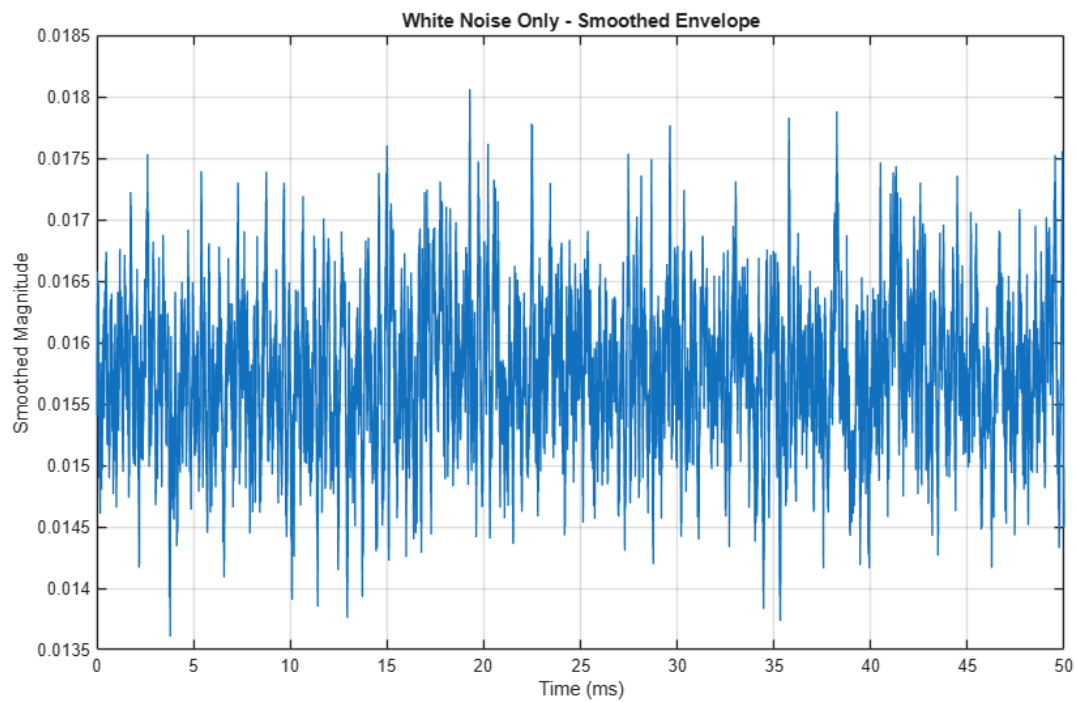
Plot Spark

```
figure;  
plot(t(1:Nshow)*1000, sparkSmooth(1:Nshow));  
xlabel('Time (ms)');  
ylabel('Smoothed Magnitude');  
title('Spark Only - Smoothed Envelope');  
grid on;
```



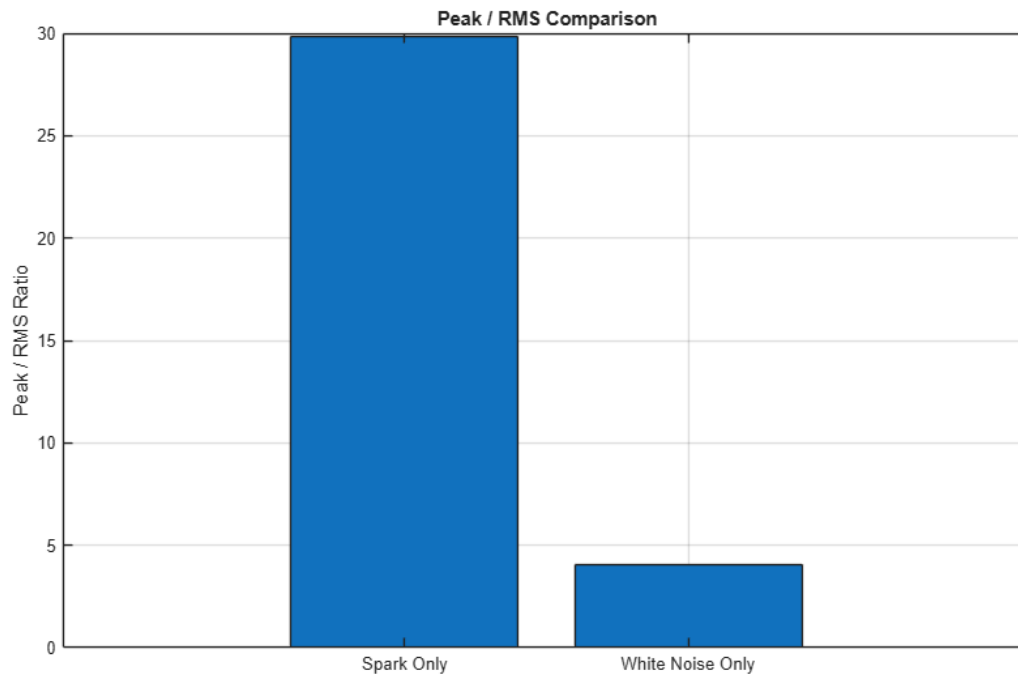
Plot White Noise

```
figure;  
plot(t(1:Nshow)*1000, noiseSmooth(1:Nshow));  
xlabel('Time (ms)');  
ylabel('Smoothed Magnitude');  
title('White Noise Only - Smoothed Envelope');  
grid on;
```



Comparison bar chart

```
figure;  
  
values = [sparkPeakRMS, noisePeakRMS];  
  
bar(values);  
  
set(gca, 'XTickLabel', {'Spark Only', 'White Noise Only'});  
  
ylabel('Peak / RMS Ratio');  
title('Peak / RMS Comparison');  
grid on;
```



Simple separation check

```
fprintf('===== COMPARISON =====\n');

if sparkPeakRMS > noisePeakRMS
    fprintf('Spark Peak/RMS is higher than white noise.\n');
    fprintf('Difference: %.2f\n', sparkPeakRMS - noisePeakRMS);
else
    fprintf('Peak/RMS does NOT separate the signals well.\n');
end

===== COMPARISON =====
Spark Peak/RMS is higher than white noise.
Difference: 25.78
```

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